

for more than 20 years, and is often offered to patients once medication no longer controls their symptoms. But despite the success of the treatment, we still do not know exactly how delivering electric pulses to brain cells creates these beneficial effects. Our results, despite being at an early-stage, suggest electric pulses boost batteries in brain cells. This potentially opens avenues for exploring how to replicate this cell powerup with non-surgical treatments, without the need for implanting electrodes in the brain.”

The early-stage study done by researchers at Imperial College London suggests the treatment boosts the number and strength of brain cell batteries called mitochondria. These batteries, in turn, provide power to brain cells, which may help reduce problems with movement and tremors.

Floating solar panels to power mines, save water

Chile is testing a floating island of solar panels as a way to generate clean energy and reduce water loss at mine operations, a cornerstone of the Andean country's economy that uses huge amounts of electricity and water. The experimental Las Tortolas power-generating island is being run by the giant Anglo-American mining company at its Los Bronces mine, and



The island of solar panels could give purpose to mine refuse in Chile by using them to generate clean energy and reduce water evaporation. (Credit: Fox News)

the initiative comes as the government pushes to put Chile at the forefront of renewable energy use in Latin America and the world.

According to *Fox News*, the 111.5sqm (1200-square-foot) array of solar panels floats in the middle of a pond, which is used to contain the refuse from mining, known as tailings. It is expected that its shadow will lower the water temperature and reduce evaporation by 80 per cent. Thus, the mine would retain more of that water for its operations and could reduce the amount of fresh water it pumps in the dry mountainous region where water is a scarce commodity.

“With this system, we can make our fresh water consump-

tion more efficient, in line with our goal of re-imagining mining and reducing Anglo-American's fresh water consumption by 50 per cent by 2030, as well as the CO₂ emissions by producing non-polluting energy,” said Patricio Chacana, Los Bronces' vice president of operations.

Tokyo unveils chatty robot-eers for 2020 Olympics

Organisers of Tokyo Olympics have rolled out a pair of chatty robots, who will be put to work on assisting wheelchair users at the 2020 Games. Referred to as distant relatives to Star Wars drone R2D2, Toyota has unveiled its human support robots (HSRs) and delivery support robots (DSRs), which are set to act as seat ushers—they will fetch food and drinks for disabled fans at next year's Olympics.



Toyota's HSR human support robot delivers a basket to a woman in a wheelchair during a demonstration of Tokyo 2020 Robot Project for the Olympic Games in Tokyo (Credit: AFP)

In an Olympic first, the waist-high, robotic volunteers will casually introduce themselves to customers, who will then be able to place orders on a tablet. “Hello, everyone! I'm a lifestyle support robot,” the fully rotating HSR said in a demonstration. “I'll fetch things for you. I also have an extendable arm that can pick things up if you drop them,” it added.

In reality, the DSR deli-bot—with its built-in drinks tray—more closely resembles a moving dustbin with blinking lights, but Toyota insisted that the technology would ensure a more comfortable Games experience for wheelchair users.

Minoru Yamauchi, general manager of 2020 robot development department, told AFP, “In the run-up to the Olympics we will work on other useful functions that these can carry out.”

Toyota is currently planning to deploy 16 units of each droid at the 2020 Olympics. Games organisers have also unveiled a Panasonic-developed exoskeleton suit to help with a range of Olympic-related chores, such as delivering or loading heavy baggage brought to Japan by athletes and officials.